Quantification: 877_wt_z081c1-2
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	14
interior lacuna count	10
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 10)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3075.75	2935.5	2349.76	px^2
major_axis_length_px	lacuna shape	88.2344	96.7141	43.2425	px
minor_axis_length_px	lacuna shape	43.1291	45.335	17.1087	px
aspect_ratio	lacuna shape	1.9916	1.9763	0.4765	unitless
eccentricity	lacuna shape	0.8367	0.8626	0.0884	unitless
solidity	lacuna shape	0.7992	0.8371	0.1049	unitless
roots_count	headline	8.2	6.5	6.2147	count
ring_length_r30_px	headline	309	282.5	208.252	px
ring_length_r60_px	reported	790.9	713.5	457.95	px
owned_length_px	ownership-dependent	1893.66	1495.52	1815.02	px
edge_count	ownership-dependent	92.2	78	83.6976	count
mean_edge_length_px	ownership-dependent	18.9956	19.7041	3.4521	px
perimeter_px	normalised	235.453	239.779	153.705	px
ring_area_r30_px2	normalised	8350.1	6108.5	4123.41	px^2
ring_area_r60_px2	normalised	20499.4	15694	8873.38	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0348711	0.0334054	0.0123318	px^-1
ring_density_r60	normalised	0.0377175	0.0409519	0.0128547	px^-1
roots_per_100px_perimeter	normalised	3.9124	3.4695	2.5064	per 100 px
ring_attached_length_r30_px	network v2	273.4	248	201.146	px
ring_attached_length_r60_px	network v2	593.2	589.5	440.042	px
ring_length_w_r30_px	network v2	352.916	318.829	241.074	px
ring_length_w_r60_px	network v2	909.382	815.648	531.073	px
sholl_crossings_r10	network v2	6.9	4.5	5.5468	count
sholl_crossings_r20	network v2	9.3	9	5.9638	count
sholl_crossings_r30	network v2	11.8	12	7.2847	count
loop_count	loops, report only	7.5	7.5	7.4274	count
loop_edge_share	loops, report only	0.24	0.2648	0.2115	share
ring_width_mean_r30_px	width	7.0838	7.0847	1.0084	px

Per field
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measure	value	units
analysed_area_px2	1.013e+06	px^2
total_skeleton_length_px	32368	px
canalicular_length_density_per_px	0.0319526	px^-1
junction_count	687	count
junction_density_per_px2	0.00067818	px^-2
lacuna_count	14	count

lacunae_per_px2	1.382e-05	px^-2
skeleton_component_count	330	count
mean_component_length_px	98.0848	px
median_component_length_px	25	px
field_length_density_w_per_px	0.0365622	px^-1
field_density_without_flagged_per_px	0.0319526	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	526	px
unattached_component_count	31	count
canalicular_length_density_attached_per_px	0.0319526	px^-1
field_length_density_with_unattached_per_px	0.0324718	px^-1
loop_count	99	count
loop_edge_share	0.2452	share
edge_count	1656	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4363	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	32368	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	828	47.9341	26.2785	1.8241	0.8363	0.811	1.1987	10.5652	544.853	True
2	4064	125.912	44.1728	2.8504	0.9364	0.8472	0.9481	38.4616	829.728	True
3	6500	135.386	68.4723	1.9772	0.8627	0.7101	0.7764	202.316	509.649	False
4	493	36.8679	18.6636	1.9754	0.8624	0.8203	-0.7656	201.503	648.073	False
5	4498	128.704	46.6689	2.7578	0.9319	0.9623	0.3797	397.582	25.7919	True
6	4895	134.554	48.0863	2.7982	0.934	0.8829	0.4584	399.457	314.021	False
7	1388	60.0833	43.9829	1.3661	0.6813	0.6761	-0.1762	427.803	598.192	False
8	1690	77.7941	42.0098	1.8518	0.8417	0.6483	0.0159	450.566	636.056	False
9	5044	117.23	57.6341	2.034	0.8708	0.8539	1.376	449.833	814.625	False
10	1085	58.3256	24.5501	2.3758	0.9071	0.9313	0.4295	634.654	16.1935	True
11	4181	115.634	46.6872	2.4768	0.9149	0.8959	1.0406	724.949	827.982	False
12	415	28.3253	19.4963	1.4529	0.7254	0.9061	-0.3467	864.431	580.06	False
13								169.697	715.862	False
14								910.754	702.905	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	225	625	1132.09	56	20.2158	7.3189
2	8	304	788	1030.69	45	22.9042	7.1897
3	20	720	1581	5700.64	259	22.0102	7.3764
4	7	314	661	795.896	53	15.0169	8.5202
5	6	247	711	1027.82	44	23.3596	8.1873

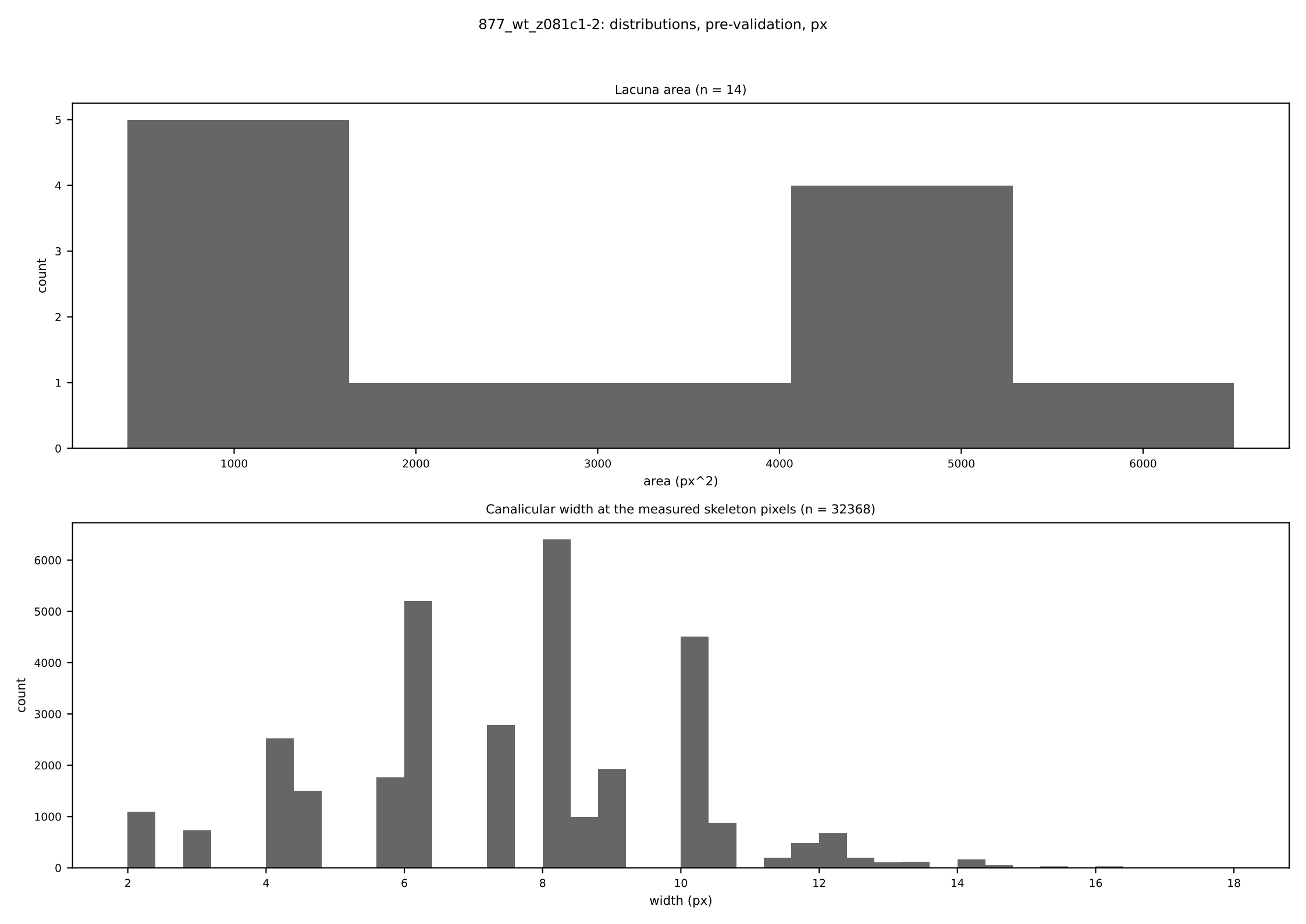
6	11	431	1176	2663.89	137	19.4444	8.0356
7	3	165	543	882.749	36	24.5208	6.5979
8	6	251	766	2108.3	103	20.4689	6.9792
9	17	531	1301	3493.66	175	19.9638	7.1902
10	0	65	236	0	0	0	7.9591
11	9	366	980	2777.99	128	21.703	6.7753
12	1	54	191	13.4853	1	13.4853	5.006
13	5	125	348	316.321	19	16.6485	8.0173
14	3	133	362	183.64	11	16.6945	6.3394

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	149.397	4144	11016	0.5976	0.5673	0.0542954	0.0567357	5.3549
2	351.404	9408	21117	0.7903	0.7218	0.0323129	0.0373159	2.2766
3	515.671	15609	34048	1	1	0.0461272	0.0464345	3.8784
4	108.397	5587	12307	1	1	0.0562019	0.0537093	6.4577
5	326.735	7997	18643	0.6576	0.6224	0.0308866	0.0381376	1.8364
6	354.617	12438	30402	1	1	0.0346519	0.0386817	3.1019
7	216.108	5237	12563	1	1	0.0315066	0.0432222	1.3882
8	263.451	6630	15255	1	1	0.0378582	0.050213	2.2775
9	355.262	12275	29427	1	1	0.0432587	0.0442111	4.7852
10	148.953	4761	12108	0.6788	0.6158	0.0136526	0.0194913	0
11	323.99	11381	28242	1	1	0.0321589	0.0347001	2.7779
12	85.9828	5167	15886	1	1	0.0104509	0.0120232	1.163
13	52.8701	4193	11362	1	1	0.0298116	0.0306284	9.4571
14	78.1838	4984	15502	1	1	0.0266854	0.0233518	3.8371

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	225	571	264.693	727.34	6	7	7
2	276	705	348.563	910.021	9	8	10
3	710	1495	833.767	1830.83	19	17	25
4	311	637	355.321	757.784	5	10	12
5	205	584	278.966	816.896	6	9	10
6	356	926	487.517	1347.01	10	15	16
7	108	314	190.409	624.399	3	6	9
8	219	587	282.338	873.512	3	8	12
9	466	944	611.5	1489.44	12	15	17
10	0	0	74.9411	270.451	0	4	4
11	277	592	416.019	1150.94	9	15	17
12	51	51	59.1127	211.48	2	1	1
13	125	220	148.924	402.576	4	4	4
14	111	166	144.255	405.848	2	2	5


877_wt_z081c1-2: distributions, pre-validation, px
Lacuna area (n = 14)
count
area (px^2)
0
1
2
3
4
5
1000
2000
3000
4000
5000
6000
Canalicular width at the measured skeleton pixels (n = 32368)
count
width (px)
0
1000
2000
3000
4000
5000
6000
2
4
6
8
10
12
14
16
18